Leetcode 151 Reverse Words in a String

Brute force

TC -> O(N)
SC -> O(N)

```
class Solution:
    def reverseWords(self, s: str) -> str:
        s= s.strip()
        words= s.split(" ")
        filtered= []
        for word in words:
            if word !="":
                filtered.append(word)
        filtered.reverse()

        return " ".join(filtered)
```

Step 1 -> strip()
Original -> " hello world"
After strip() : "hello world"

Step 2 -> split(" ")
words = s.split(" ")
["hello", "", "", "world"]

Step 3 -> Remove empty strings

word	filtered
"hello"	["hello"]
""	["hello"]
""	["hello"]
"world"	["hello", "world"]

filtered = ["hello", "world"]

Step -4 Reverse()
["world", "hello"]

Step 5 → Join
"world", "hello"]

```python
class Solution:
    def reverseWords(self, s: str) -> str:
        def clean_spaces(s):
            left, right = 0 , len(s)-1
            while left <= right and s[left] == " ":
                left+=1
            while left <=right and s[right]== " ":
                right-=1

            result=[]
            while left<=right:
                if s[left] != " ":
                    result.append(s[left])
                elif result[-1] != " ":
                    result.append(s[left])
                left+=1
            return "".join(result)

        def reverse(lst, l,r):
            while l <r:
                lst[l],lst[r]=lst[r],lst[l]
                l+=1
                r-=1


        def reverse_each_word(lst):
            n= len(lst)
            start=0
            end=0
            while start < n:
                while end < n and lst[end] != " ":
                    end+=1
                reverse(lst , start ,end-1)
                start = end+1
                end+=1
        s= clean_spaces(s)
        lst= list(s)
        reverse(lst, 0 , len(lst)-1)
        reverse_each_word(lst)
        return "".join(lst)
```

Optimal In Place

TC → O(N)

S(→ O(N)

S = " hello world "

After Removing heading/ Trailing Spaces
"hello world"

Then collasping multiple Spaces manually
" hello world"

Step2 → Convert to list
lst = ['h', 'e', 'l', 'l', 'o', ' ', 'w', 'o', 'r', 'l', 'd']

Step3 → Reverse Entire String
reverse (lst, 0, len(lst) - 1)
['d', 'l', 'r', 'o', 'w', ' ', 'o', 'l', 'l', 'e', 'h']
↳ 'dlrow olleh'

Step4 → Reverse Each word
first word → 'dlrow' → 'world'
Second word → 'olleh' → 'hello'
Final List = ['w', 'o', 'r', 'l', 'd', ' ', 'h', 'e', 'l', 'l', 'o']
"world hello"